

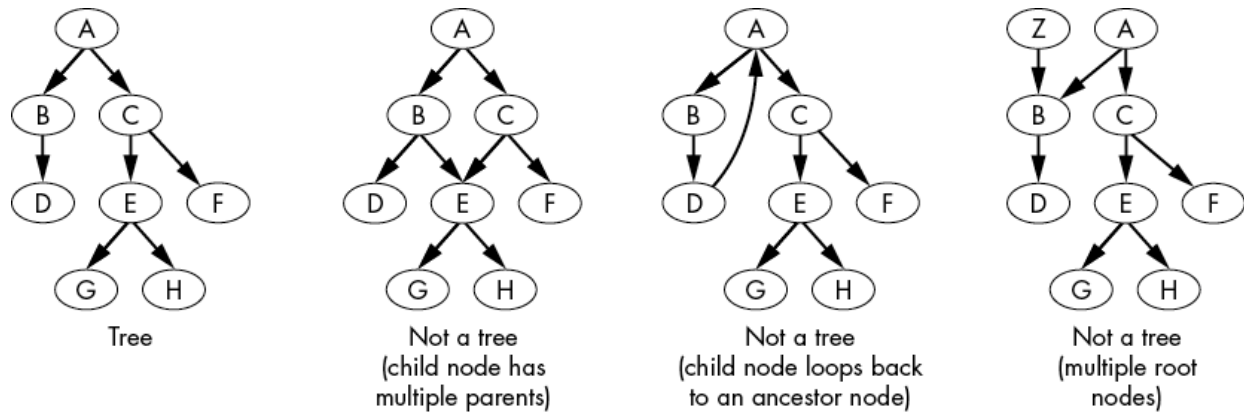


Figure 4-1: A tree (left) and three examples of non-trees

As you can see, child nodes must have one parent and not have an edge that creates a loop, or else the structure is no longer considered a tree. The recursive algorithms we cover in this chapter apply only to tree data structures.

A Tree Data Structure in Python and JavaScript

Tree data structures are often drawn growing downward, with the root at the top. [Figure 4-2](#) shows a tree created with the following Python code (it's also valid JavaScript code):

```

root = {'data': 'A', 'children': []}
node2 = {'data': 'B', 'children': []}
node3 = {'data': 'C', 'children': []}
node4 = {'data': 'D', 'children': []}
node5 = {'data': 'E', 'children': []}
node6 = {'data': 'F', 'children': []}
node7 = {'data': 'G', 'children': []}
node8 = {'data': 'H', 'children': []}
root['children'] = [node2, node3]
node2['children'] = [node4]
node3['children'] = [node5, node6]
node5['children'] = [node7, node8]

```
